Automatic Vehicle Gate Access System in All-Weather Conditions

License Plate Recognition + Vehicle Make & Model Recognition

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B.S. Computer Engineering — FYP 2025

Problem & Motivation

Manual gate checks are slow and costly. Camera-based access works in clear weather, but fog, rain, glare, and night degrade plate reading and vehicle identification. Plate-only systems are vulnerable to copied plates mounted on the wrong vehicle.

Objectives

- Restore weather-degraded frames before inference
- Detect and read plates with two-stage YOLOv8n
- Classify make & model with EfficientNet-B4 (245 classes)
- Fuse plate + vehicle checks in a dual-verification policy
- Demonstrate fog, rain, and night branches on real imagery

Dual Verification

GRANT — plate and vehicle match whitelist
DENY — no matching record
ALERT — plate matches but vehicle disagrees (cloned-plate scenario)

Hardware Target

IR lane camera, **Orange Pi 5** (8 GB) edge node, opto-isolated relay, commercial gate motor controller, outdoor PSU, IP-rated enclosure.

System Pipeline

Camera frame I
↓
Weather mitigation (fog / rain / night)
↓
YOLOv8n plate detection
↓
YOLOv8n character OCR (38 classes → string s)
↓
EfficientNet-B4 VMMR (245 classes → label y)
↓
Policy $\mathcal{D}(s, y, \mathcal{W})$ GRANT / DENY / ALERT

Weather Preprocessing

Fog/haze: dark-channel prior, guided filtering, LAB CLAHE
Rain: vertical median stack + Telea inpainting (plate edges protected)
Night: adaptive gamma + tone lift

Technology Stack

Python, PyTorch, OpenCV, Ultralytics YOLOv8n, EfficientNet-B4, Node.js dashboard, Colab/CUDA training.

Key Results

Plate det. mAP@0.5
98.07%

Character OCR mAP@0.5*
99.4%

VMMR Top-1 accuracy
96.24%

VMMR Top-5 accuracy
99.39%

*Character validation mirrored training pool (Roboflow v4 export lacked a held-out split). LPD and VMMR metrics use standard disjoint test splits.

Detailed Metrics

LPD (epoch 50)	
Precision	98.50%
Recall	95.53%
mAP@0.5:0.95	70.73%
OCR (epoch 100)	
Precision	99.0%
Recall	96.8%
F1	97.9%
VMMR (17,254 test)	
Macro F1	96.18%

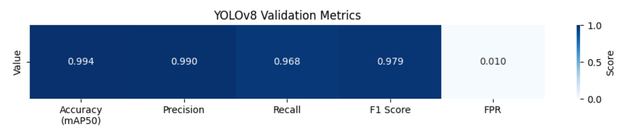
Datasets

Roboflow LPD v1: 23,531 images
Roboflow OCR v4: 1,235 crops, 38 classes
VMMR corpus: 57,135 images, 245 classes

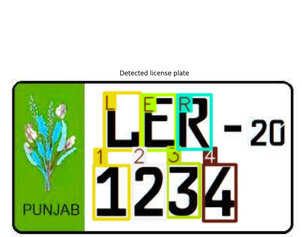
Qualitative Results



Fog branch: restoration + LPD + OCR + VMMR overlays



YOLOv8n character validation metrics heatmap





Character boxes (left) and VMMR inference overlay (right)

Contributions & Future Work

Done: end-to-end weather-aware pipeline; dual-check policy; reproducible archived metrics; web dashboard prototype.
Next: Orange Pi latency profiling; unified inference resolution; string-level benchmarks per weather branch; field relay deployment.

Keywords: ANPR • VMMR • YOLO • EfficientNet • Dark Channel Prior • Access Control • Intelligent Transportation Systems